ELEC2134 : Circuits and Signals

PART A (Week 1 to Week 5)

Transform Methods

- Periodic signals and Fourier series
- Aperiodic signals and Fourier transforms
- Laplace transforms of signals and circuits

System Analysis

- Linear systems and convolution
- Impulse response
- Frequency response and transfer functions
- Steady-state analysis of linear systems
- Transient analysis of 1st and 2nd systems
- Circuit analysis using transform methods

PART B (Week 7 to Week 10)

AC Circuit Analysis

- AC circuits with sinusoidal inputs
- Use of phasors and complex impedance in AC circuit analysis
- AC power: real, reactive and apparent power
- Power factor (leading/lagging)
- **Series and parallel resonance**
- Transformers and coupled coils
- Application of network theorems: Two port networks

Adapted by Aous Naman (PhD) from lectures by Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

ELEC2134 : Circuits and Signals (PART B)

Topic 8: AC Resonance

- ✓ Series Resonance
- ✓ Parallel Resonance
- ✓ Series to Parallel Conversion

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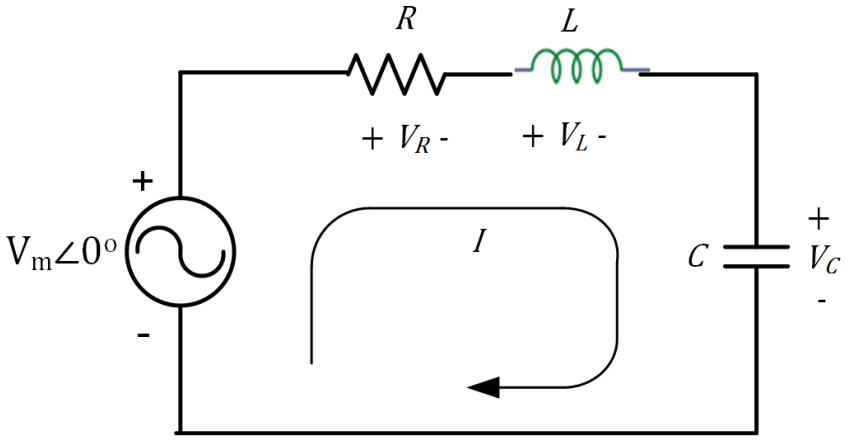
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Resonance

- ✓ Resonance is the tendency of a system to oscillate at maximum amplitude at a certain frequency known as the resonant frequency.
- ✓ The possibility of resonance always exists wherever there is periodic motion - movement that is repeated at regular intervals.
- ✓ The resonant system may be electrical, mechanical, hydraulic , acoustic or some other kind.
- ✓ Electrical resonance occurs in an electric circuit at a particular frequency when the imaginary parts of impedances of circuit elements cancel each other.
- ✓ A circuit is in resonance when the voltage and current at the circuit input terminals are in phase.
- ✓ Resonant circuits can generate higher voltages and currents than the input voltage or current.

Series Resonance

- ✓ RLC circuits posses the distinctive ability to provide natural responses of the oscillatory type. The energy flows back and forth between the capacitance and the inductance.
- ✓ Consider a series RLC circuit



- ✓ The impedance seen by the source is

$$Z = R + j\omega L + \frac{1}{j\omega C} = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

$$Z = R + jX \quad \text{where} \quad X = \omega L - \frac{1}{\omega C}$$

← net reactance

- ✓ Depending on the frequency ω of the applied source, we have three possibilities

$$X = \omega L - \frac{1}{\omega C} \qquad Z = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

(i) $\omega L < \frac{1}{\omega C}$ or $X < 0$ $\Rightarrow$ indicating capacitive behaviour

(ii) $\omega L > \frac{1}{\omega C}$ or $X > 0$ $\Rightarrow$ indicating inductive behaviour

(iii) $\omega L = \frac{1}{\omega C}$ or $X = 0$

$Z = R + j(0)$

$Z = R \leftarrow$ Resistive

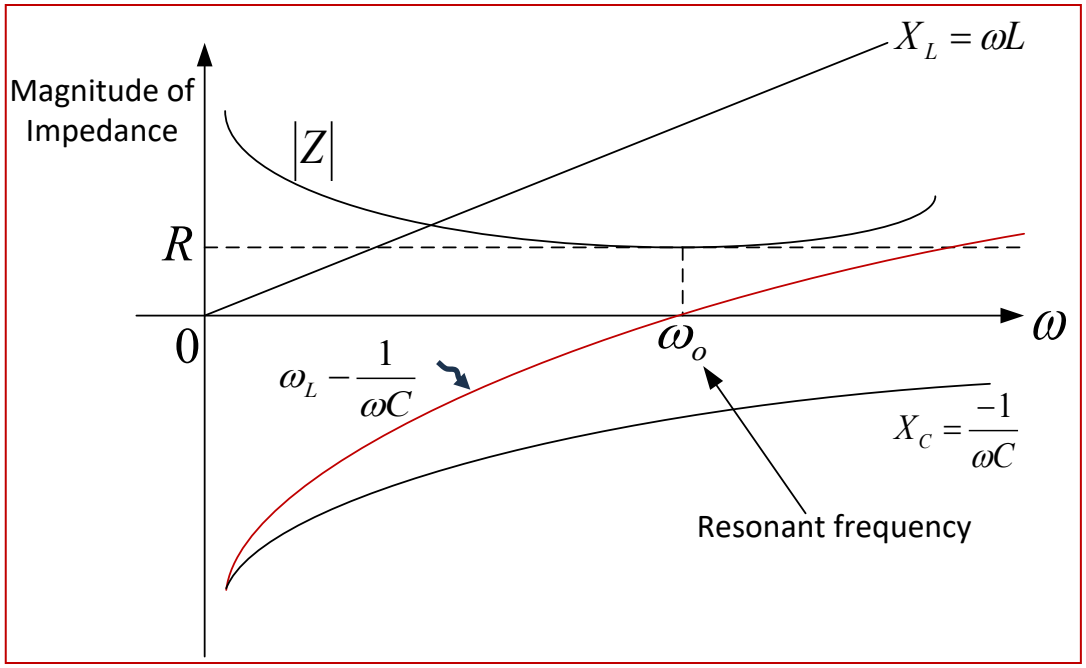
$\Rightarrow$ The RLC circuit behaves resistively, with I and V in phase with each other.

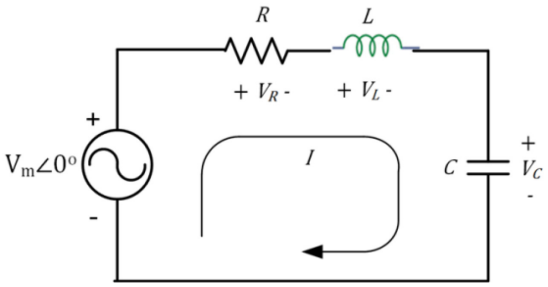
- ✓ This behaviour, referred to as unity power-factor resonance, occurs at the special frequency ω_o that makes $\omega_o L = \frac{1}{\omega_o C}$ or $\omega_o^2 = \frac{1}{LC}$.

$$\omega_o = \frac{1}{\sqrt{LC}} \text{ rad/sec} \quad \Rightarrow \quad f_o = \frac{1}{2\pi\sqrt{LC}} \text{ Hz}$$

ω_o is called the resonance frequency

- ✓ For obvious reasons, ω_o is called the resonance frequency. Even though the individual impedances Z_L and Z_C are not zero, at resonance frequency they cancel each other out, making the 'LC' combination act as a short circuit.

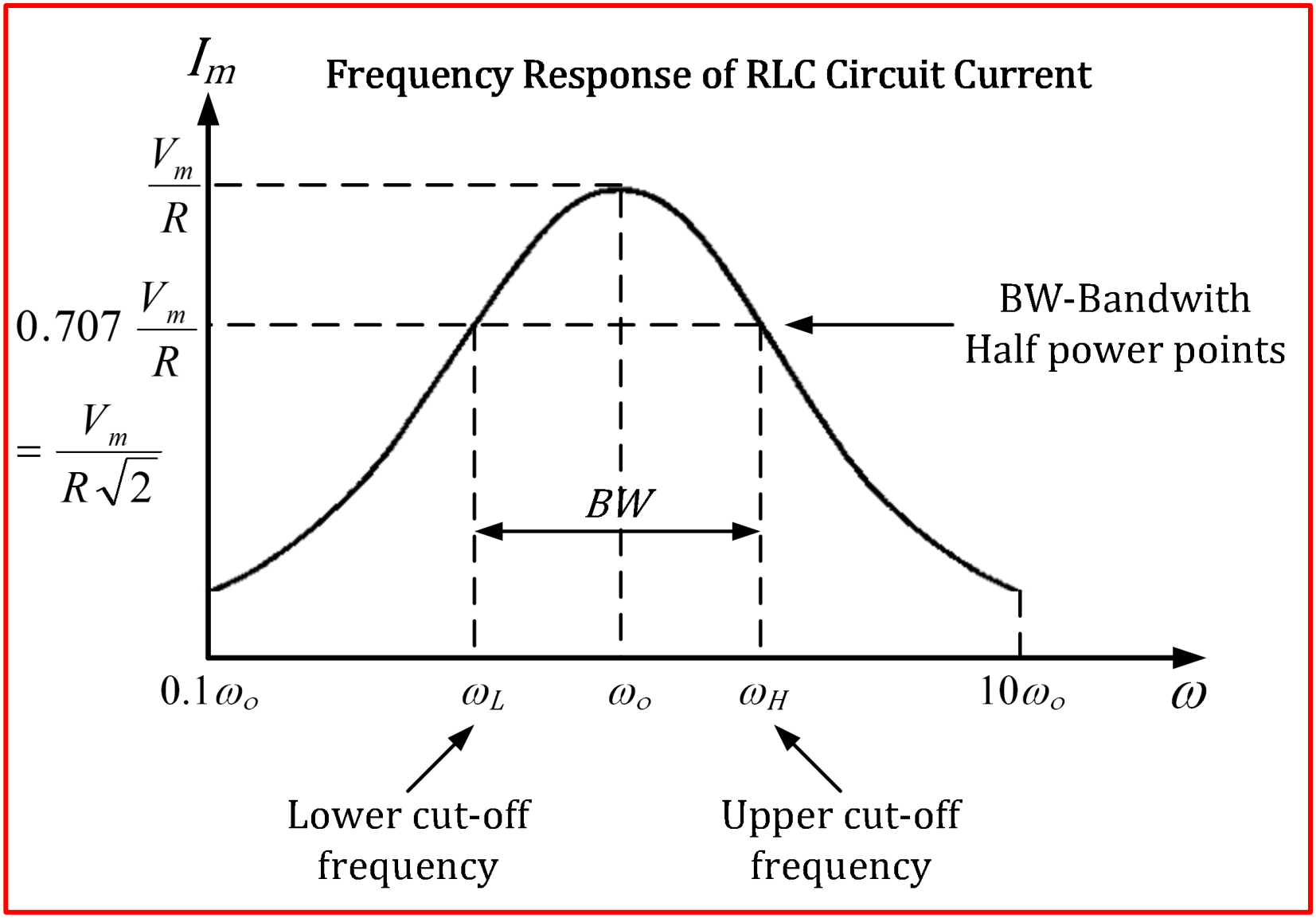




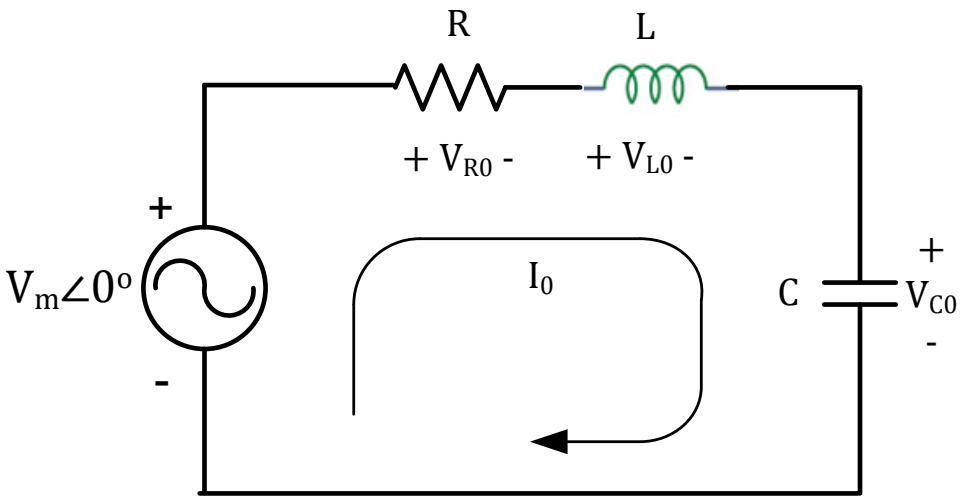
- ✓ The component of Z in a series RLC circuit is

$$Z = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

✓ Figure below shows the current at resonance



At Resonance



V_{LO}: Voltage across the inductor at resonance

V_{CO}: Voltage across the capacitor at resonance

$$\omega_o L = \frac{1}{\omega_o C} \Rightarrow \omega_o = \frac{1}{\sqrt{LC}}$$

$$I_o = \frac{V_m}{R}$$

Resonant frequency

Current at resonance

$$V_{Lo} = (\omega_o L) I_o = \omega_o L \frac{V_m}{R} = \frac{\omega_o L}{R} V_m$$

$$V_{Co} = \left(\frac{1}{\omega_o C}\right) I_o = \frac{1}{\omega_o C} \frac{V_m}{R} = \frac{1}{\omega_o CR} V_m$$

At resonance

At series resonance $V_{Lo} = V_{Co}$ (equal impedance)

At Resonance

At series resonance $V_{Lo} = V_{Co}$ (equal impedance)

$$\therefore \frac{\omega_o L}{R} = \frac{1}{\omega_o RC}$$

$$V_{Lo} = \frac{\omega_o L}{R} V_m$$

$$V_{Co} = \frac{1}{\omega_o CR} V_m$$

✓ Then by definition, the quality factor of the series RLC circuit

$$Q = \frac{\omega_o L}{R} = \frac{1}{\omega_o RC}$$

✓ We have $\omega_o = \frac{1}{\sqrt{LC}} \Rightarrow \therefore Q = \frac{\omega_o L}{R} = \frac{1}{R\sqrt{\frac{C}{L}}}$

✓ At resonance $\omega = \omega_o$, voltage across the inductor at resonance,

$$V_{Lo} = \frac{\omega_o L}{R} V_m = QV_m \longrightarrow \boxed{V_{Lo} = QV_m}$$

✓ At resonance $\omega = \omega_o$, voltage across the capacitor at resonance,

$$V_{Co} = \frac{1}{\omega_o CR} V_m = QV_m \longrightarrow \boxed{V_{Co} = QV_m}$$

The peak voltage amplitude across the reactive elements (at resonance) exceeds the peak amplitude of the applied signal (V_m).

Resonance Voltage Rise

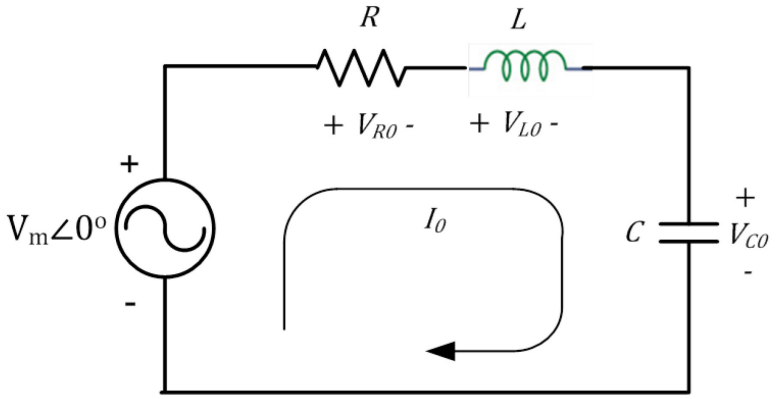
- ✓ At resonance

$$I = \frac{V_m}{R}\angle 0°$$

$$V_{Ro} = V_m\angle 0°$$

$$V_{Lo} = QV_m\angle 90°$$

$$V_{Co} = QV_m\angle -90°$$



$$V_{Lo} = V_{Co} \text{ and } Z = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

= 0

$$\therefore Z = R$$

Note: Maximum current occurs in the circuit at resonance

- ✓ Clearly if $Q > 1$, the peak voltage across the reactive elements will be greater than that of the applied voltage, a phenomenon known as resonance voltage rise.

Resonance Voltage Rise

- ✓ The amount of magnification depends on the quality factor.
- ✓ For example if $Q = 100$, $V_m = 10\angle 0°$, then

$$V_{Lo} = 100 \times 10 = 1000\ V$$

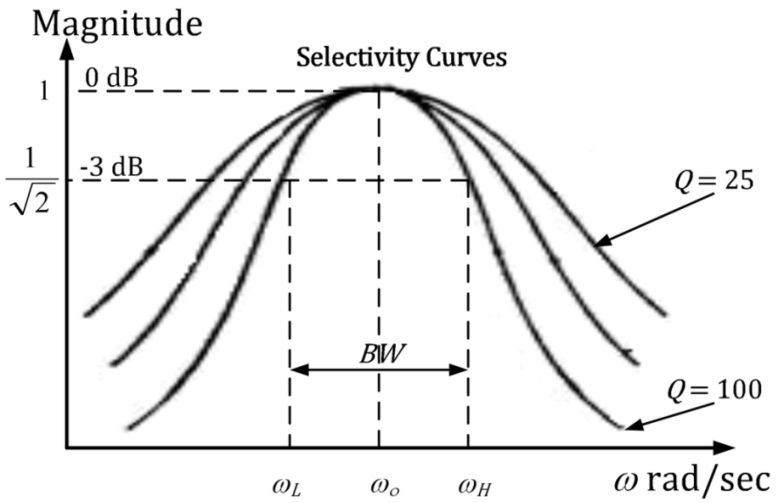
$$V_{Co} = 100 \times 10 = 1000\ V$$

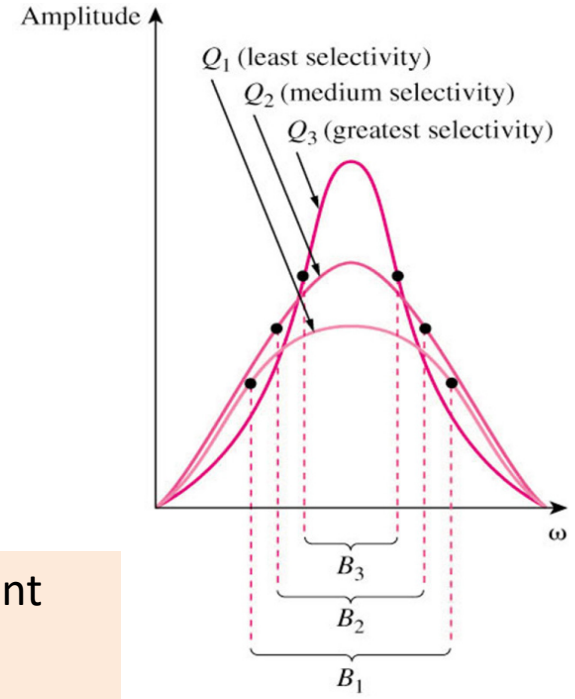
Much greater than the applied voltage

- ✓ It is interesting to note that the smaller the resistance R in a series RLC circuit, the higher the value of Q and hence greater amount of voltage rise at resonance.
- ✓ In the limit $R \to 0$, we would have $Q = \frac{\omega_o L}{R} = \frac{1}{\omega_o CR} \to \infty$ and this would result in an infinite voltage rise.
- ✓ The quality factor is also a measure of frequency selectivity. Thus, we say that a circuit with a high Q has a high selectivity, whereas a low Q circuit has low selectivity.

- ✓ Figure below shows the relative response versus ω for $Q = 25$, 50, and 100 where we observe that highest Q provides the best frequency selectivity, i.e. the higher rejection of signal component outside the bandwidth $BW (= \omega_U - \omega_L)$ which is the difference in 3 dB frequencies.

$$Bandwidth\ (BW) = \omega_U - \omega_L$$





The quality factor (Q) is the ratio of its resonant frequency to its bandwidth. $\boldsymbol{Q = \frac{\omega_0}{BW} = \frac{R}{L}}$

$$\boldsymbol{Z} = R + j\left(\omega L - \frac{1}{\omega C}\right) = R\left[1 + j\frac{\omega L}{R} - j\frac{1}{\omega CR}\right]$$

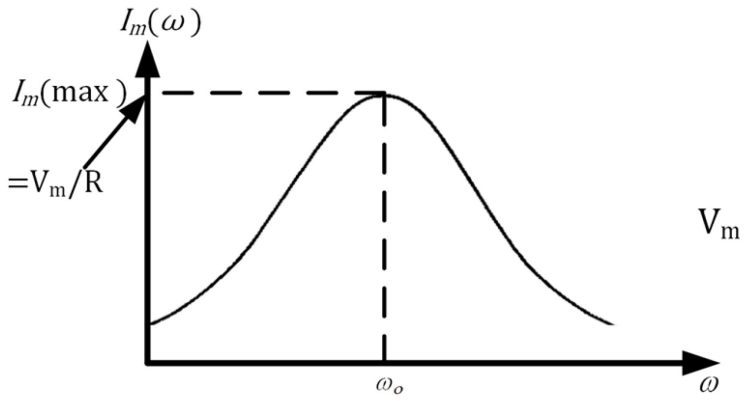
$$\boldsymbol{Z} = R\left[1 + jQ\left(\frac{\omega}{\omega_o} - \frac{\omega_o}{\omega}\right)\right]$$

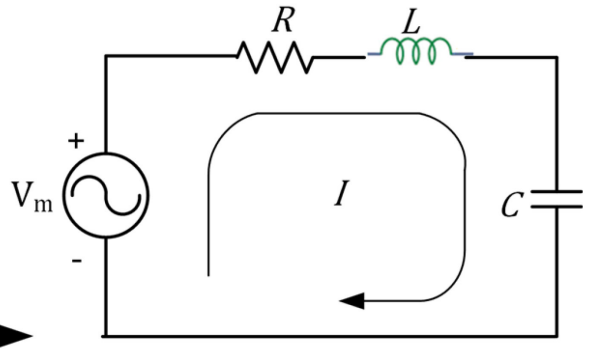
If $\omega_o = \frac{1}{\sqrt{LC}}$ and $Q = \frac{\omega_o L}{R} = \frac{1}{\omega_o CR}$, then $\frac{L}{R} = \frac{Q}{\omega_o}$ and $\frac{1}{CR} = Q\omega_o$

$$I_m = \frac{V_m \angle 0°}{\boldsymbol{Z}}$$

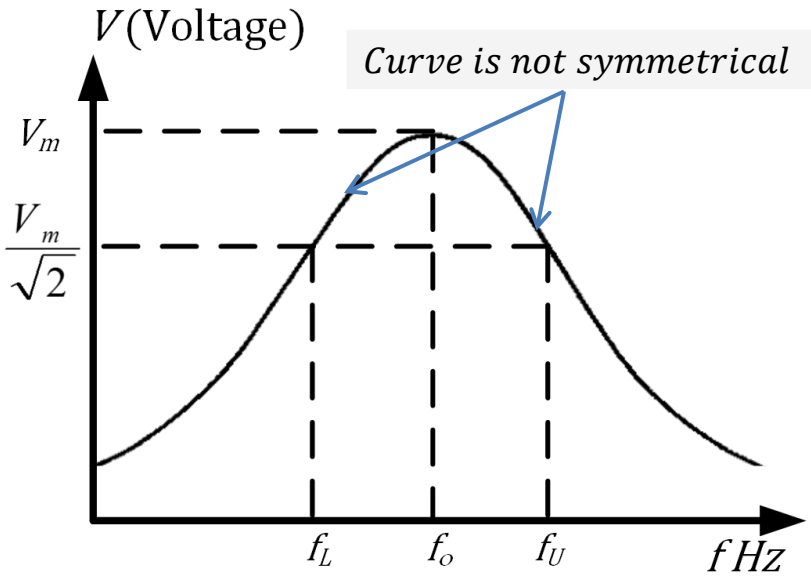
$$I_m(\omega) = \frac{V_m}{R}\frac{1}{\sqrt{1 + Q^2\left(\frac{\omega}{\omega_o} - \frac{\omega_o}{\omega}\right)^2}}$$

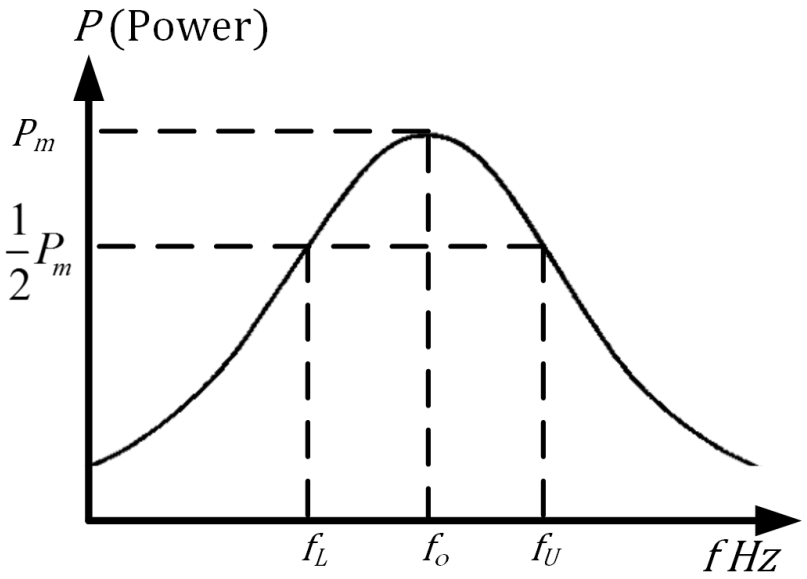
- ✓ The plot of I_m versus ω for a given V_m has the familiar bell shaped profile of the band-pass function.





- ✓ At resonance, the average power P dissipated in the resistance is also maximised, $P_{\max} = \frac{\left(\frac{V_m}{\sqrt{2}}\right)^2}{R} = \frac{V_m^2}{2R}$
- ✓ The frequencies at which P is down to half its maximum value are the half-power frequencies ω_L and ω_U.
- ✓ Half-power $BW = \omega_U - \omega_L$





- ✓ We can show that, $Q = \frac{Resonant\ Frequency\ (f_0)}{Bandwidth\ (BW)}$
- ✓ If $Q \gg 1$, then the half-power frequencies are given by the approximate expressions:

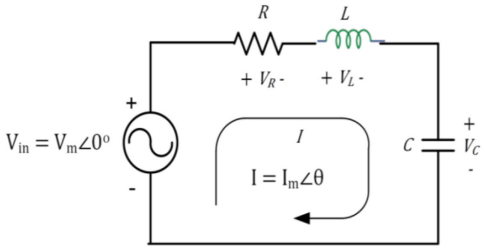
$$f_U \approx f_o + \frac{BW}{2}$$

$$f_L \approx f_o - \frac{BW}{2}$$

Example 1

- Let $V_{in} = 10\angle 0°$, $R = 400\ \Omega$, $L = 10mH$, $C = 10nF$. Calculate the phasor voltages across the elements (if ω_o is the resonant frequency) for

  (a) $\omega = \frac{\omega_o}{2}$ (b) $\omega = \omega_o$ (c) $\omega = 2\omega_o$



✓We have $\omega_o = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10\times10^{-3}\times10\times10^{-9}}} = 10^5\ rad/sec$

(a) $\omega = \frac{\omega_0}{2} = \frac{10^5}{2}$ rad/sec

$$\boldsymbol{I} = \frac{V_{in}}{R + j\left(\omega L - \frac{1}{\omega C}\right)} = \frac{10\angle 0°}{400 + j\left(\frac{10^5}{2}\times 10\times 10^{-3} - \frac{1}{\frac{10^5}{2}\times 10\times 10^{-9}}\right)} = 6.442\angle 75.07°\ mA$$

$$\boldsymbol{V}_R = 400\times 6.442\angle 75.07° = 2.577\angle 75.07°$$

$$\boldsymbol{V}_L = j\omega L\times I = \frac{10^5}{2}\times 10\times 10^{-3}\times 6.442\angle(75.07° + 90°) = 3.221\angle 165.07°$$

$$V_C = \frac{1}{j\omega C}\ \mathrm{x}\ I = \frac{1}{\frac{10^5}{2}\mathrm{x}\ 10\ \mathrm{x}\ 10^{-9}}\mathrm{x}\ 6.442\angle(75.07° - 90°) = 12.884\angle -14.93°$$

(b) $\omega = \omega_o = 10^5\ rad/sec$, we obtain

$$I_m = \frac{V_m \angle 0^\circ}{R} = \frac{10\angle 0^\circ}{400} \Rightarrow \therefore I = 25\angle 0^\circ\ mA$$

$$V_R = 400 \text{ x } 25 \text{ x } 10^{-3} = 10\ V$$

$$V_L = j\omega L \times I = 25\angle 90^\circ$$

$$V_C = \frac{1}{j\omega C} \times I = 25 \angle -90^\circ$$

$$V_L = V_C$$

(c) $\omega = 2\omega_o = 2 \times 10^5\ rad/sec$, we likewise obtain

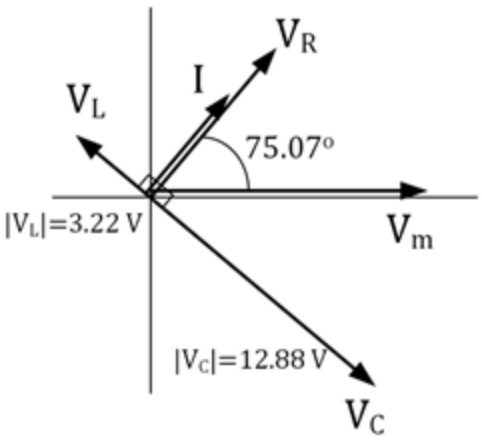
$$I = 6.442\angle -75.07^\circ$$

$$V_R = 2.577\angle -75.07^\circ$$

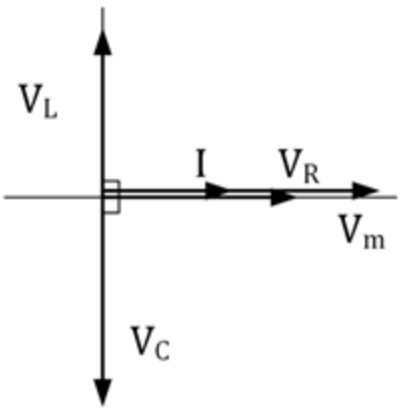
$$V_L = 12.88\angle 14.93^\circ$$

$$V_C = 3.221 \angle -165.07^\circ$$

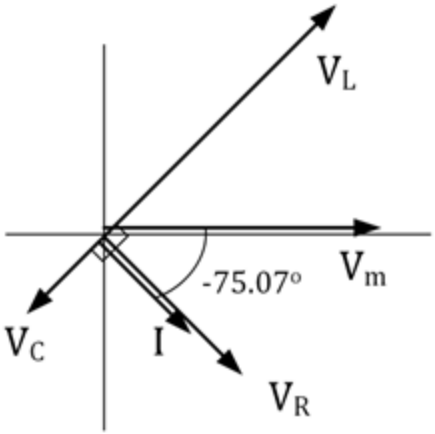
Phasor Diagrams



$\omega = \frac{\omega_o}{2}$



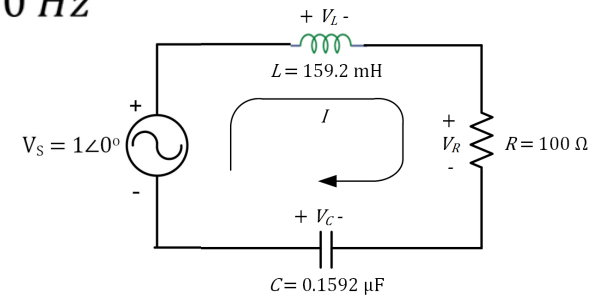
$\omega = \omega_o$



$\omega = 2\omega_o$

Example 2

- A series resonant circuit is shown below. Calculate the resonant frequency, the bandwidth and the half-power frequencies. Assuming that the frequency of the source is the same as the resonant frequency, find the phasor voltages across the elements and draw a phasor diagram.



- ✓ Resonant frequency, $f_o = \dfrac{1}{2\pi\sqrt{0.1592\times0.1592\times10^{-6}}} = 1000\ Hz$
- ✓ Quality factor, $Q = \dfrac{2\pi f_o L}{R} = \dfrac{2\pi\times1000\times0.1592}{100} = 10$
- ✓ Bandwidth, $BW = \dfrac{f_o}{Q} = \dfrac{1000}{10} = 100\ Hz$
- ✓ Approximate half power frequencies

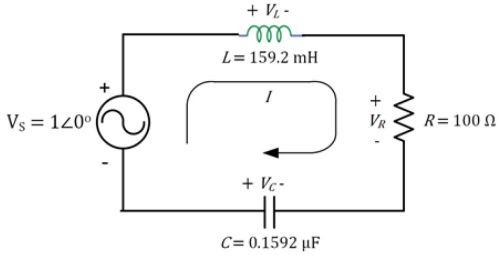
$$f_U \approx f_o + \frac{BW}{2} = 1000 + \frac{100}{2} = 1050\ Hz$$

$$f_L \approx f_o - \frac{BW}{2} = 1000 - \frac{100}{2} = 950\ Hz$$

$$Z_L = j2\pi f_o L = j2\pi \times 1000 \times 0.1592 = j1000\ \Omega$$

$$Z_C = \frac{1}{j2\pi f_o C} = -\frac{j}{2\pi \times 1000 \times 0.1592 \times 10^{-6}} = -j1000\ \Omega$$

- ✓ As expected, the reactances are equal in magnitude at the resonant frequency.



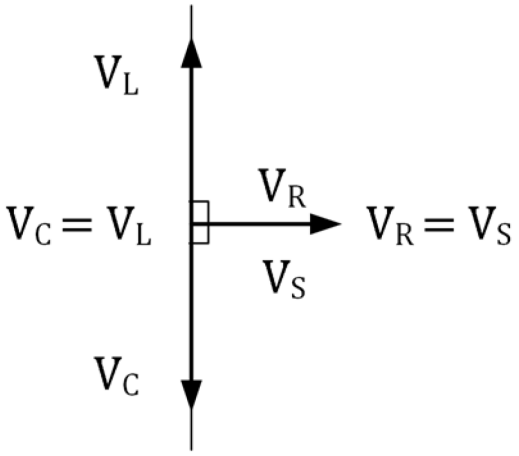
- ✓ The total impedance of the circuit

$$Z_T = R + Z_L + Z_C$$
$$= 100 + j1000 - j1000 = 100\Omega$$

- ✓ Phasor current $I = \frac{V_S}{Z_T} = \frac{1\angle 0°}{100} = 0.01\angle 0°$
- ✓ Voltage across the elements

$$V_R = RI = 100\times 0.01\angle 0° = 1\angle 0°$$
$$V_L = Z_L I = j1000\times 0.01\angle 0° = 10\angle 90°$$
$$V_C = Z_C I = -j1000\times 0.01\angle 0° = 10\angle -90°$$

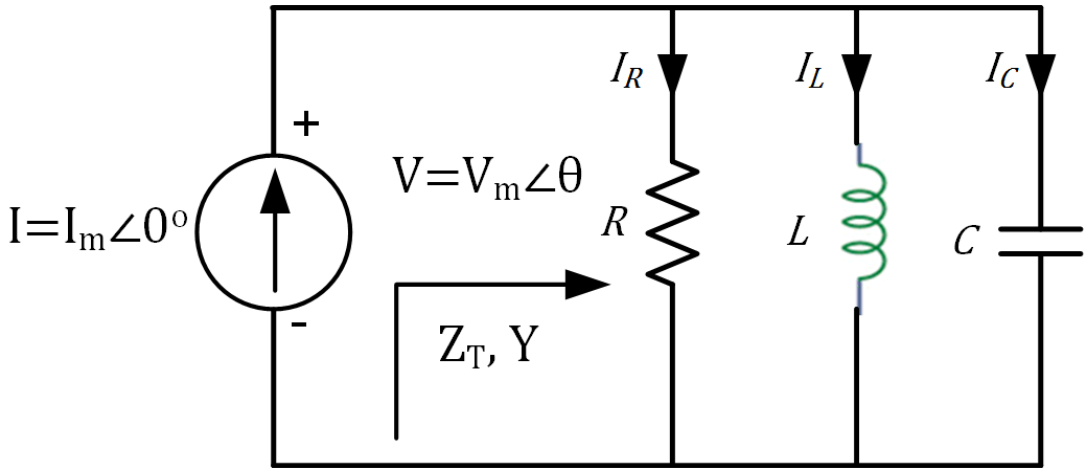
Phasor Diagrams



Notice that the voltage across the inductance and capacitance are much larger than the source voltage in magnitude; Nevertheless, Kirchoff's voltage law is satisfied because V_L and V_C are out of phase and cancel.

Parallel Resonance

- ✓ Another type of circuit known as a parallel resonant circuit is shown below:



- ✓ The admittance seen by the source is the sum of the three admittances.

$$Y = Y_R + Y_C + Y_L$$

$$Y = G + j\omega C + \frac{1}{j\omega L}$$

$$Y = G + j\left(\omega C - \frac{1}{\omega L}\right)$$

- ✓ We expect a resonance behaviour when the susceptance ωC and $\frac{-1}{\omega L}$ cancel each other out and this occurs when $\omega = \omega_o$. Hence, we have $$\omega_o C = \frac{1}{\omega_o L}$$
- ✓ ∴The resonance frequency $$\omega_o = \frac{1}{\sqrt{LC}}$$
- ✓ $\omega_o = \frac{1}{\sqrt{LC}}$ is exactly the same as the expression for the resonant frequency of the series circuit.
- ✓ We observe that at parallel resonant frequency

$$Y = G$$

$$Z_T = \frac{1}{G} = \frac{1}{Y} = R \quad \text{(resistive)}$$

and $\theta = 0°$

$$Y = G + j\left(\omega C - \frac{1}{\omega L}\right)$$

✓ At resonance

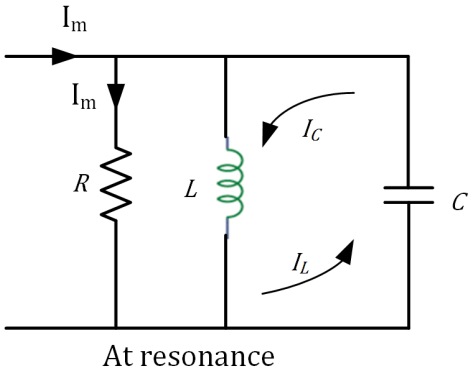
$V = V_m \angle 0°$ $\qquad V_m = I_m \times R$ (magnitude)

$$|I_C| = \frac{V_m}{\frac{1}{\omega_o C}} = V_m \omega_o C = I_m (R \omega_o C) \quad \text{(magnitude)}$$

$$|I_L| = \frac{V_m}{\omega_o L} = \frac{I_m R}{\omega_o L} = I_m \left(\frac{R}{\omega_o L}\right) \quad \text{(magnitude)}$$

✓ At parallel resonance,

$$|I_L| = |I_C| \Rightarrow R\omega_o C = \frac{R}{\omega_o L}$$



At resonance

✓ By definition quality factor at parallel resonance

$$Q = R\omega_o C = \frac{R}{\omega_o L}$$

$$\boldsymbol{Y} = G + j\left(\omega C - \frac{1}{\omega L}\right) = G\left[1 + j\left(\omega CR - \frac{R}{\omega L}\right)\right]$$

If $R = \frac{1}{G}$, $\omega_o = \frac{1}{\sqrt{LC}}$, $Q = R\omega_o C = \frac{R}{\omega_o L}$, then

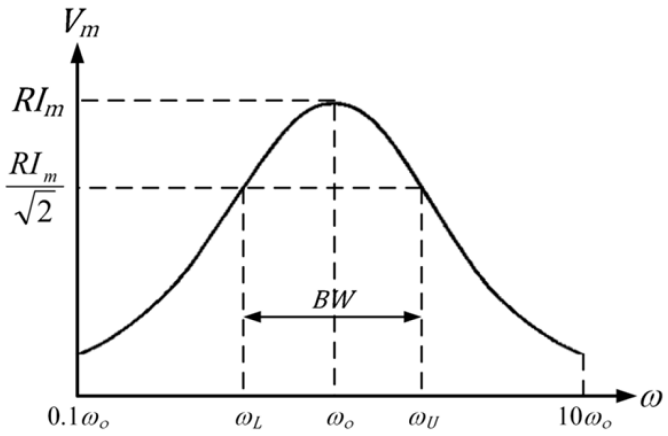
$$\boldsymbol{Y} = G\left[1 + jQ\left(\frac{\omega}{\omega_o} - \frac{\omega_o}{\omega}\right)\right]$$

$\boldsymbol{V} = V_m\angle\theta$ and $\boldsymbol{I} = Im\angle 0°$

$$\boldsymbol{V} = Z\boldsymbol{I} = \frac{1}{Y}\boldsymbol{I} = \frac{I_m\angle 0°}{G\left[1+jQ\left(\frac{\omega}{\omega_o}-\frac{\omega_o}{\omega}\right)\right]} \rightarrow$$

$$V_m(\omega) = RI_m \frac{1}{\sqrt{1 + Q^2\left(\frac{\omega}{\omega_o} - \frac{\omega_o}{\omega}\right)^2}}$$

✓ The plot $V_m(\omega)$ versus ω is shown below. Taking magnitude,



Peak value of $V_m(max) = RI_m$ at $\omega = \omega_o$, $\omega_o C = \frac{1}{\omega_o L}$ making Y purely conductive, $Y = G + j(0)$

- ✓ Eventhough the individual admittance Y_L and Y_C are nonzero, their sum vanishes at resonance, making the parallel combination of L and C effectively behave as an open circuit.
- ✓ ∴ The average power dissipated in the resistance

$$P_{\max} = \left(\frac{I_m}{\sqrt{2}}\right)^2 R = RI_{rms}^2$$

- ✓ The frequencies at which P is down to half its maximum value are the half power frequencies ω_L and ω_U and the half-power $BW = \omega_U - \omega_L$, such that

$$Q = \frac{\omega_o}{BW}$$

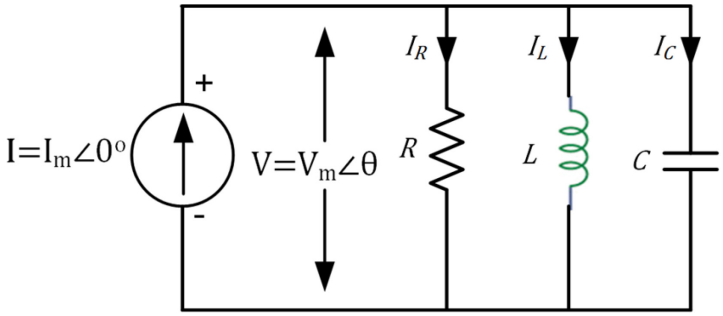
$$Q = R\omega_o C = \frac{R}{\omega_o L}$$

Combining these equations, we get

$$BW = \frac{1}{RC}$$

Example 3

- Find the L and C values for a parallel resonant circuit that has $R = 10\ k\Omega$, $f_o = 1\ MHz$, $BW = 100\ kHz$. If $I = 10^{-3}\angle 0°$, draw the phasor diagram showing the currents through each of the elements in the circuit at resonance.



✓ $Q = \frac{f_o}{BW} = \frac{10^6}{10^5} = 10$

✓ $L = \frac{R}{2\pi f_o Q} = \frac{10^4}{2\pi \times 10^6 \times 10} = 159.2\ \mu H$

✓ $C = \frac{Q}{2\pi f_o R} = \frac{10}{2\pi \times 10^6 \times 10^4} = 159.2\ pF$

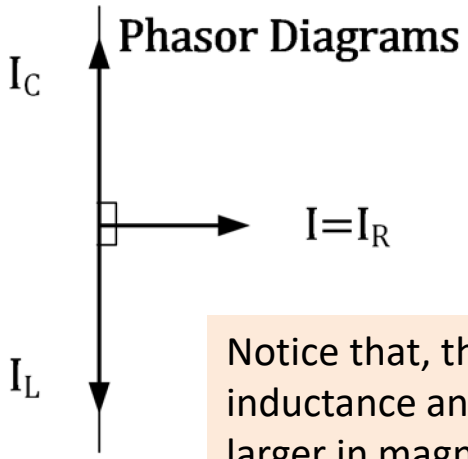
✓At resonance

$$V = IR = 10^{-3}\angle 0° \times 10^4 = 10\angle 0°$$

$$I_R = \frac{V}{R} = \frac{10\angle 0°}{10^4} = 10^{-3}\angle 0°$$

$$I_L = \frac{V}{j2\pi f_o L} = \frac{10\angle 0°}{j10^3} = 10^{-2}\angle -90°$$

$$I_C = \frac{V}{\frac{-j}{2\pi f_o C}} = \frac{10\angle 0°}{-j10^3} = 10^{-2}\angle 90°$$

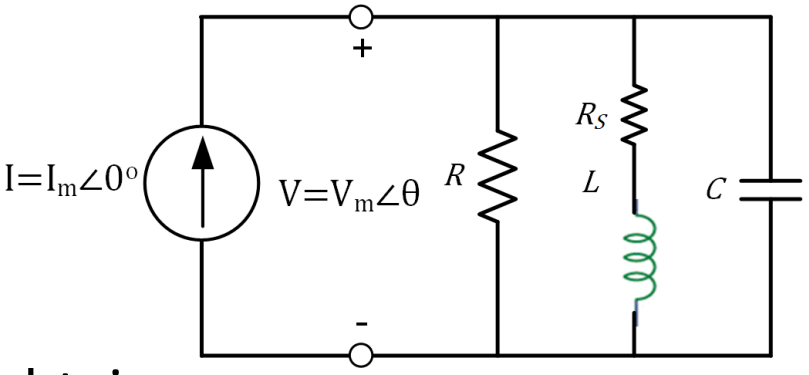


Notice that, the current through the inductance and capacitance are larger in magnitude than the applied source current. However, since I_C and I_L are out of phase, they cancel.

Practical Resonant Circuit

- ✓ The parallel RLC configuration is known as tuned circuit, such as tuned oscillators and tuned amplifiers in radio communications.
- ✓ Of the three components making up an RLC circuit, the inductor is the least ideal because of the winding resistance (R_S).
- ✓ We wish to investigate the impact of R_S upon the resonance frequency.
- ✓ For an ideal inductor $R_S = 0$
- ✓ For non ideal inductor, we have

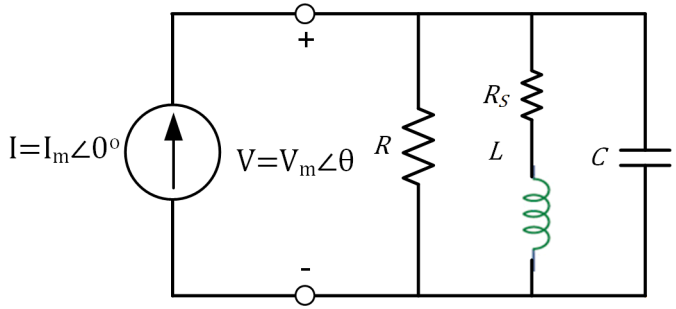
$$Y = \frac{1}{R} + j\omega C + \frac{1}{R_S + j\omega L}$$



- ✓ By rationalising the last term, we obtain
- ✓ $$Y = \frac{1}{R} + \frac{R_S}{R_S^2 + (\omega L)^2} + j\omega\left(C - \frac{L}{R_S^2 + (\omega L)^2}\right)$$

- ✓ $Y = \frac{1}{R} + \frac{R_S}{R_S^2+(\omega L)^2} + j\omega\left(C - \frac{L}{R_S^2+(\omega L)^2}\right)$
- ✓ We are interested in the frequency that makes Y purely real
- ✓ $\therefore$ Imaginary part = 0

$$C - \frac{L}{R_S^2+(\omega L)^2} = 0 \;\Rightarrow\; \therefore \;\; \omega_o = \sqrt{\frac{1}{LC} - \frac{R_S^2}{L^2}}$$



- ✓ In the limit $R_S \to 0$ (ideal case)

$$\omega_o \to \frac{1}{\sqrt{LC}}$$

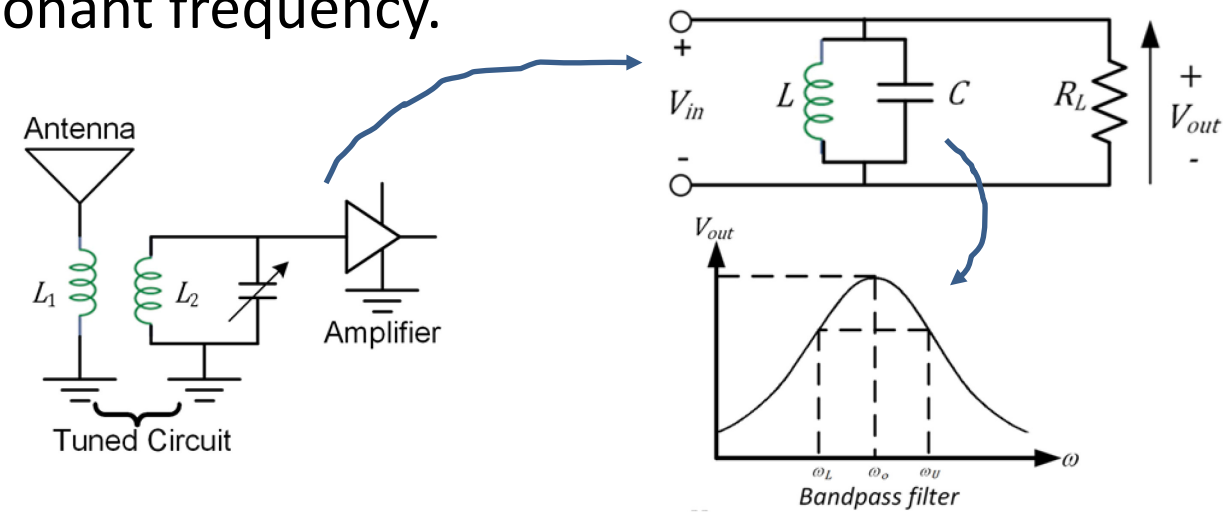
- ✓ The quality factor of the coils is

$$Q_{coil} = \frac{\omega_o L}{R_s}$$

$$\therefore \omega_o = \frac{1}{\sqrt{LC}}\left[\sqrt{1 - \frac{1}{Q_{coil}^2}}\right]$$

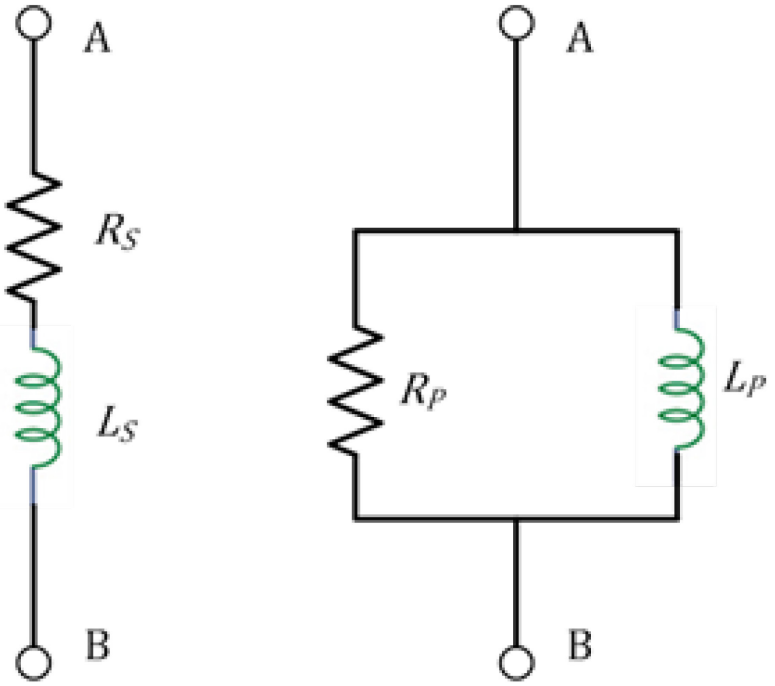
- ✓ Clearly, the higher the value of Q_{coil} the ω_o is close to $\frac{1}{\sqrt{LC}}$

- ✓ When a radio or TV receivers is tuned to a particular station or a channel, it is set to operate at the resonant frequency of that station.
- ✓ A parallel circuit has high impedance at its resonant frequency. Therefore it attenuates signals at all frequencies except the resonant frequency.



- ✓ Varying the capacitance of the tuned circuit, will change the resonant frequency. Generally inductance is kept constant and the capacitor value is changed as we select different stations or channels.

Series to Parallel Conversion



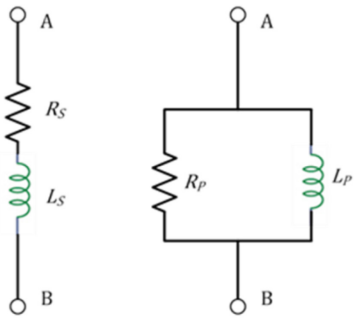
Series to parallel conversion of an RL circuit

- ✓ It is possible to select R_p and L_p so that, at a given frequency ω, the admittance of the parallel structure equals that of the series structure.

$$\frac{1}{R_p} + \frac{1}{j\omega L_p} = \frac{1}{R_s + j\omega L_s}$$

✓ Rationalising both sides and equating the real and imaginary parts yields

$$\frac{1}{R_p} - j\frac{1}{\omega L_p} = \frac{R_s}{R_s^2+\omega^2 L_s^2} - j\frac{\omega L_s}{R_s^2+\omega^2 L_s^2}$$



Equating the real parts

$$\therefore \frac{1}{R_p} = \frac{R_s}{R_s^2+\omega^2 L_s^2} \rightarrow \therefore R_p = \frac{R_s^2+\omega^2 L_s^2}{R_s}$$

If $R_s \downarrow$ then $R_p \uparrow$

R_P = Parallel resistance

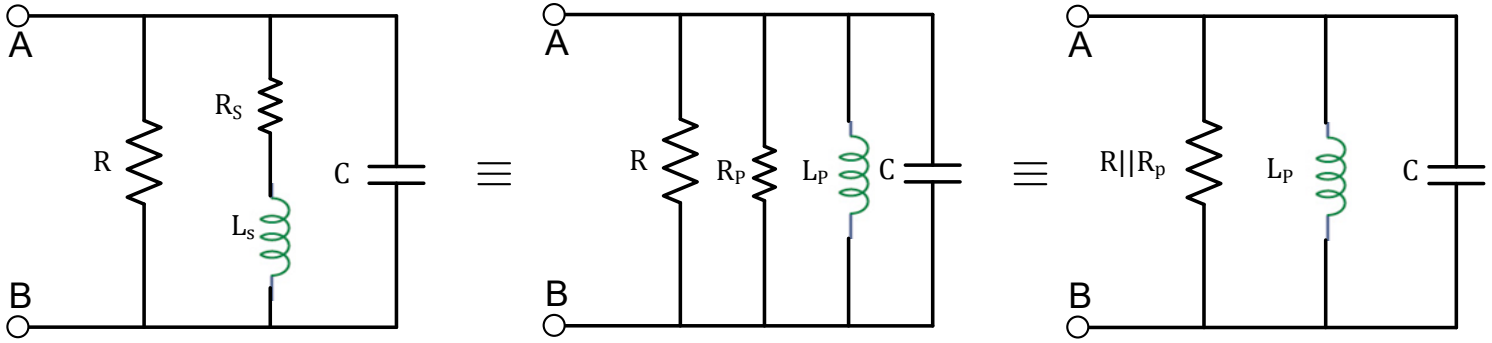
Equating the imaginary parts

$$\therefore \frac{-1}{\omega L_p} = \frac{-\omega L_s}{R_s^2+\omega^2 L_s^2} \rightarrow \therefore L_p = \frac{R_s^2+\omega^2 L_s^2}{\omega^2 L_s}$$

If R_s is very small $L_p \approx L_s$

Note: $X_p = \omega L_p = \omega\left(\frac{R_s^2+\omega^2 L_s^2}{\omega^2 L_s}\right) = \frac{R_s^2+(\omega L_s)^2}{\omega L_s}$

✓ The parallel RLC configuration finds application in a class of circuits known as tuned circuits, tuned amplifiers and tuned oscillators.



$$Q_{coil} = \frac{\omega_o L_S}{R_S}$$

Small coil resistance provides large Q_{coil}

$$\therefore \omega_o = \frac{1}{\sqrt{L_P C}} \Leftarrow resonant\ frequency$$

Small R_S implies large R_P.

$$R_P = \frac{R_S^2 + \omega L_S^2}{R_S}$$

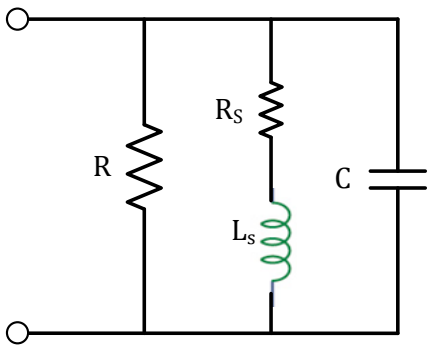
Small R_S implies $L_P \approx L_S$

$$L_p = \frac{R_s^2 + \omega^2 L_s^2}{\omega^2 L_s}$$

$$Q_{circuit} = R_p||R \times \omega_o \times C = \frac{R_P||R}{\omega_o L_P}$$

Example 4

- A parallel RLC circuit has $R = 100\,k\Omega$, $L_s = 10\,mH$, $C = 10\,nF$. Find resonant frequency ω_o, bandwidth BW, and the quality factor $Q_{circuit} = \frac{\omega_o}{BW}$ for (a) $Q_{coil} = \infty$ and (b) $Q_{coil} = 50$.



✓ (a) With $Q_{coil} \to \infty$ i.e. $R_S \to 0$ $\quad Q_{coil} = \frac{\omega_o L_S}{R_S}$

$$\omega_o = \frac{1}{\sqrt{L_S C}} = \frac{1}{\sqrt{10\times10^{-3}\times10\times10^{-9}}} = 100000\,rad/sec$$

$$BW = \frac{1}{RC} = \frac{1}{100\times10^{3}\times10\times10^{-9}} = 1000\,rad/sec$$

$$Q_{circuit} = \omega_o RC = \frac{100000}{1000} = 100$$

$$Q_{circuit} = R\omega_o C = \frac{R}{\omega_o L}$$

$$Q_{circuit} = \frac{\omega_o}{BW}$$

$$BW = \frac{1}{RC}$$

✓ (b) With $Q_{coil} = 50$ i.e. $R_S = \frac{\omega_o L_S}{Q_{coil}}$

$$L_P = \frac{R_S^2 + \omega_o^2 L_S^2}{\omega_o^2 L_S}$$

$R_S = \frac{10^5 \times 10 \times 10^{-3}}{50} = 20\ \Omega;$ $\quad \therefore \omega_o = \frac{1}{\sqrt{L_P C}} = \frac{10^5}{\sqrt{1+\frac{1}{50^2}}} = 99980$ rad/sec

$$\omega_o = \frac{1}{\sqrt{L_S C}} \frac{1}{\sqrt{1 + \frac{1}{Q_{coil}^2}}}$$

$$BW = \frac{1}{\{R||[R_S(1 + Q_{coil}^2)]\}C}$$

$$= \frac{1}{\{10^5||20(1 + 50^2)\} \times 10 \times 10^{-9}} = 2999\ rad/sec$$

$$Q_{circuit} = \frac{99980}{2999} = 33.33$$

$$Q_{circuit} = \frac{\omega_o}{BW}$$

Note : The quality factor of the circuit has changed from 100 (when $Q_{coil} = \infty$ or $R_s = 0$) to 33.33 (when $Q_{coil} = 50$ or $R_S = 20\ \Omega$)

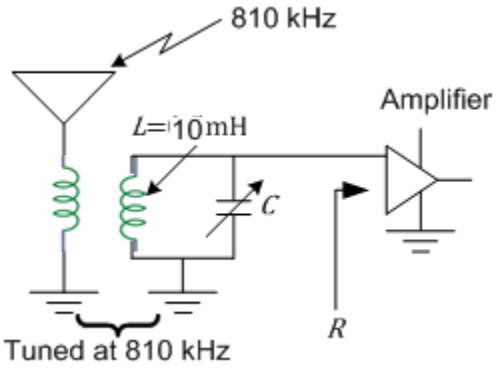
Summary

	Series Resonance	Parallel Resonance
Equation	$f_o = \frac{1}{2\pi\sqrt{LC}}$	$f_o = \frac{1}{2\pi\sqrt{LC}}$
Circuit current	Max current: $\frac{V_S}{R_S}$	Max voltage: $V_m = I_m R$
Bandwidth	$BW = \frac{f_o}{Q}$	$BW = \frac{f_o}{Q}$
Quality factor	$Q = \frac{\omega_o L}{R_S} = \frac{1}{\omega_o C R_S}$	$Q = \omega_o C R = \frac{R}{\omega_o L}$
Resonant rise	Voltage (V_C or V_L)$= QV_S$	Current (I_C or I_L)$= QI_m$
Phase angle	$\angle\theta = 0°$	$\angle\theta = 0°$
Power factor	$PF = \cos 0° = 1$	$PF = \cos 0° = 1$

Example 5

- A radio receiver with a parallel RLC circuit has an inductance $L = 10\ mH$. It is tuned to a radio station transmitting at 810 kHz frequency.

  (a) What is the value of the capacitor of the circuit at this resonant frequency?



$$\omega_o = \frac{1}{\sqrt{LC}} \leftarrow \text{resonant frequency}$$

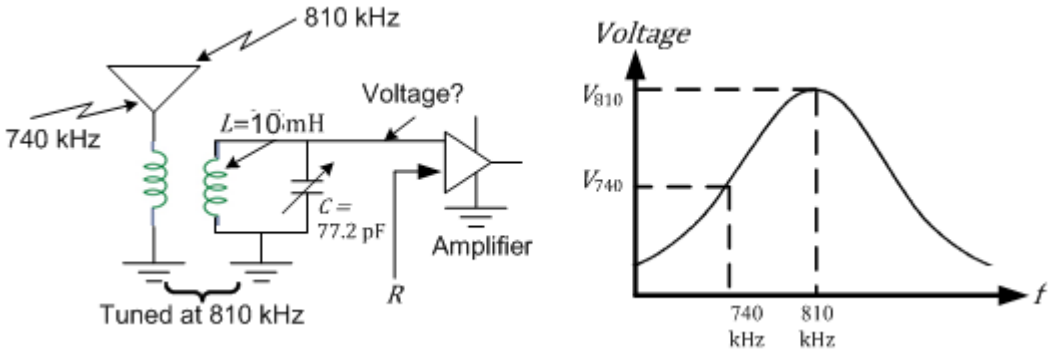
$$\therefore f_o^2 = \frac{1}{4\pi^2 LC}$$

$$\therefore C = \frac{1}{4\pi^2 \times 0.5 \times 10^{-3} \times (810 \times 10^3)^2} = 77.2\ pF$$

(b) What is the value of resistance R if $Q = 75$?

$$Q = \omega_o CR \Rightarrow \therefore R = \frac{75}{2\pi \times 810 \times 10^3 \times 77.2 \times 10^{-12}} = 190.89\ k\Omega$$

(c) If a nearby radio station transmits 740 kHz and both signals are picked up by the antenna having the same current amplitude $I\ \mu A$, what is the ratio of the voltage at 810 kHz to the voltage at 740 kHz.



$$\left|\frac{V_{810HZ}}{V_{740HZ}}\right| = ? \qquad |V_{810HZ}| = IR = I\times 190.89\times 10^3$$

$$Y = \frac{1}{R} + \frac{1}{\frac{1}{j\omega C}} + \frac{1}{j\omega L} = \frac{1}{R} + j\omega C - \frac{j}{\omega L}$$

$$|Y| = \sqrt{\left(\frac{1}{R}\right)^2 + \left(\omega C - \frac{1}{\omega L}\right)^2} = 78.18\ \mu\Omega^{-1}$$

$\omega = 740\ kHz$
$R = 190.89\ k\Omega$
$C = 77.2\ pF$
$L = 10\ mH$

$$|V_{740}| = \frac{I}{|Y|} \Rightarrow \therefore \left|\frac{V_{810kHZ}}{V_{740kHZ}}\right| = \frac{I\times 190.89\times 10^3}{I/78.18\ ?\Omega^{-1}} = 14.9$$

That is, the voltage developed across the parallel circuit when it is tuned at f=810 kHz is 14.9 times larger than the voltage developed at f=740 kHz.

ELEC2134: Circuits and Signals

AC Circuit Analysis

Tutorial questions

AC Resonance

- ✓ Series Resonance
- ✓ Parallel Resonance
- ✓ Series to Parallel Conversion

Adapted by Aous Naman (PhD) from lectures by
Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

Tutorial Questions (AC Resonance)

Question 1 (Pass-Level): Determine the R and C values for a series resonant circuit that has $L = 10\mu H$, $f_o = 1\ MHz$, and $Q = 50$. Find the bandwidth and approximate half-power frequencies of the circuit.

Answer: $C = 2533\ pF$, $R = 1.257\ \Omega$, $BW = 20\ kHz$, $f_L \approx 990\ kHz$, and $f_U \approx 1010\ kHz$

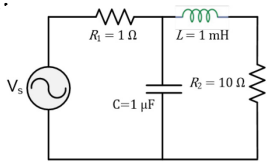
Question 2 (Pass-Level): (a) Using a $100\mu H$ inductance, specify suitable values for R and C to implement a series RLC circuit that resonates at 1 MHz with a quality factor of 100.

Answer: (a) R = 6.283 Ω and $C = 253.3\ pF$,

(b) If the circuit is driven with a source having a peak amplitude of 1 Volt, find the power it dissipates at resonance, as well as the amplitudes of the voltages across its reactive elements.

Answer: $79.58\ mW$, $100\ V$

Question 3 (Pass-Level): For the circuit shown below $V_s = 170\cos\omega t$ and $Q = 50$. Find the resonance frequency ω_o, bandwidth BW, upper cut-off frequency ω_U and the lower cut-off frequency ω_L.



Ans: $\omega_o = 30000$ rad/sec,
$BW = 600\ \frac{rad}{\text{sec}}$
$\omega_L = 29700$ rad/sec,
$\omega_U = 30300\ rad/sec$

Question 4 (Pass-Level): Find the R and L values for a series resonant circuit that has C = 470 pF, a resonant frequency of 5 MHz, and a bandwidth of 200 kHz.

Answer: (a) R = 2.709 Ω and L= 2.156 μH,

Question 5 (Pass-Level): A parallel resonant circuit has $f_0 = 10\ MHz$, $BW = 20\ kHz$ and $R = 1k\Omega$. Find L and C.

Answer: L= 0.3183 μH, C= 795.8 pF

Question 6 (Higher–Level): The circuit shown in Fig 6.1 can be converted to Fig. 6.2. Using Fig 6.2 ,

(i) show that $\omega_o = \frac{1}{\sqrt{LC}}\frac{1}{\sqrt{1+\frac{1}{Q_{coil}^2}}}$ where $Q_{coil} = \frac{\omega_o L}{R}$

(ii) Show that the Bandwidth (BW) $BW = \frac{1}{\{R /\!/ [R_s(1+Q_{coil}^2)]\}C}$

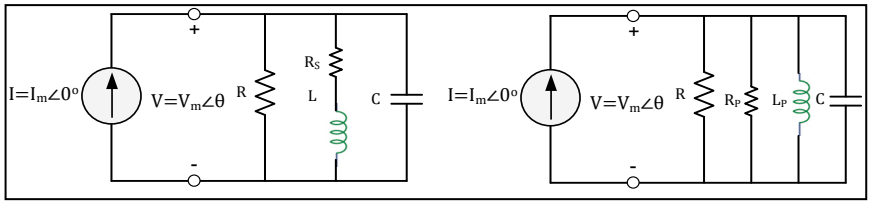


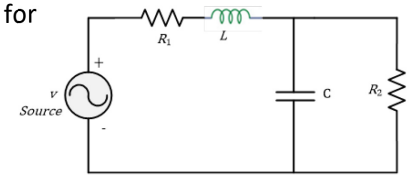
Fig 6.1 Fig 6.2

Question 7 (Higher–Level): Show that provided $R_2 > \sqrt{\frac{L}{C}}$ the impedance $Z(\omega)$ seen by the source of figure below becomes purely resistive for

$\omega = \omega_o = \left(\frac{1}{\sqrt{LC}}\right)\sqrt{1-\frac{L}{R_2^2 C}}$

that Z at $\omega_o = R_1 + \frac{L}{R_2 C}$

(ω_o- resonant frequency)



Question 8 (Higher–Level): A series resonant circuit is shown below. The half-power frequencies (ω_u and ω_l) can be obtained by setting the impedance |Z| equal to $\sqrt{2}\,R$. Show that:

1. $\omega_l = \frac{-R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \left(\frac{1}{LC}\right)}$
1. $\omega_u = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \left(\frac{1}{LC}\right)}$
2. $BW = \frac{R}{L}$

